ELEC70092
EE9-CS5-2
EE9-FPN2-08

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

SYSTEMS IDENTIFICATION AND LEARNING

Monday, 12 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Parisini
Second Marker(s) : F. Teng

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SYSTEMS IDENTIFICATION AND LEARNING

1. Consider the stochastic process $y(t)$ generated by the following state equations:

$$\begin{cases} x_1(t+1) = 0.3x_1(t) + 0.5x_2(t) + v(t+1) \\ x_2(t+1) = 0.2x_1(t) - 0.1x_2(t) \\ y(t) = x_1(t) \end{cases}$$

where $v(\cdot) \sim WN(0,1)$ is a white process.

a) Determine, if possible, a stationary ARMA model for $y(t)$.

[4 Marks]

b) Write the difference equation for the optimal one-step-ahead linear predictor $\widehat{y}(t+1|t)$ of $y(t+1)$ based on past measurements $\{y(t), y(t-1), ...\}$.

[4 Marks]

c) Compute the prediction error $\varepsilon_1(t) = y(t+1) - \widehat{y}(t+1|t)$ in terms of samples of the white process $v(\cdot)$. Moreover, compute the variance $\text{var}[\varepsilon_1(t)]$ of the prediction error $\varepsilon_1(t)$.

[3 Marks]

d) Similarly, write the difference equation for the optimal two-step-ahead predictor $\widehat{y}(t+2|t)$ of $y(t+2)$ based on past measurements $\{y(t), y(t-1), ...\}$.

[3 Marks]

e) Compute the prediction error $\varepsilon_2(t) = y(t+2) - \widehat{y}(t+2|t)$ in terms of samples of the white process $v(\cdot)$. Moreover, compute the variance $\text{var}[\varepsilon_2(t)]$ of the prediction error $\varepsilon_2(t)$.

[3 Marks]

f) Compare the two prediction error variances $\text{var}[\varepsilon_1(t)]$ and $\text{var}[\varepsilon_2(t)]$ and discuss your findings.

[3 Marks]

2. Consider the zero-mean stochastic process $y(\cdot)$ described by the following dynamic model:

$$y(t) = \frac{8 + 18z^{-1} + 4z^{-2}}{4 + 2z^{-1}}\omega(t) + \frac{2 + 5z^{-1}}{4 + z^{-1}}\eta(t),$$

where $\omega(\cdot) \sim WN(0, 0.2)$ and $\eta(\cdot) \sim WN(0, 3)$ are zero-mean mutually uncorrelated white processes.

a) Show that the stochastic process $y(\cdot)$ is stationary and compute its variance $\text{var}[y(t)]$.

[7 Marks]

b) Consider the zero-mean stochastic process $\widetilde{\omega}(\cdot)$ defined as:

$$\widetilde{\omega}(t) = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^k e(t-k),$$

where $e(\cdot) \sim WN(0, 0.7)$ is a white process uncorrelated with $\eta(\cdot)$ and consider the stochastic process $\widetilde{y}(\cdot)$ described by the following dynamic model:

$$\widetilde{y}(t) = \frac{8 + 18z^{-1} + 4z^{-2}}{4 + 2z^{-1}}\widetilde{\omega}(t) + \frac{2 + 5z^{-1}}{4 + z^{-1}}\eta(t)$$

i) Determine the correlation function $\gamma_{\widetilde{\omega}}(\tau)$ of the process $\widetilde{\omega}(\cdot)$.

[7 Marks]

ii) Compute the variance $\text{var}[\widetilde{y}(t)]$ of the process $\widetilde{y}(\cdot)$.

[6 Marks]

3. Suppose that a large number N of measurements $\{v(1), v(2), ..., v(N)\}$ of a stationary discrete-time stochastic process $v(\cdot)$ are available.

a) Consider the family of stochastic models $\mathscr{M}_1(b_1)$:

$$v(t) = b_1 v(t-1) + \varepsilon_1(t),\ \varepsilon_1(t) \sim WN(0, \sigma_1^2)$$

Devise a least-squares procedure to compute the estimates $\widehat{b}_1$ and $\widehat{\sigma}_1^2$ of the parameter b_1 and the noise variance σ_1^2, respectively.

[6 Marks]

b) Consider the family of stochastic models $\mathscr{M}_2(a_1, a_2)$:

$$v(t) = a_1 v(t-1) + a_2 v(t-2) + \varepsilon_2(t),\ \varepsilon_2(t) \sim WN(0, \sigma_2^2)$$

Devise a least-squares procedure to compute the estimates $\widehat{a}_1$, $\widehat{a}_2$, and $\widehat{\sigma}_2^2$ of the parameters a_1, a_2, and the noise variance σ_2^2, respectively.

[7 Marks]

c) Consider the family of stochastic models $\mathscr{M}_3(c_1)$:

$$v(t) = c_1 u(t-1) + \eta(t),\ \eta(t) \sim WN(0, \sigma_3^2)$$

where $\{u(t)\}$ denotes a **known** exogenous input. Devise a least-squares procedure to compute the estimates $\widehat{c}_1$ and $\widehat{\sigma}_3^2$ of the parameter c_1 and the noise variance σ_3^2, respectively.

[7 Marks]

4. The **unknown** and **constant** temperature of a chemical reactor is denoted by T. Three independent sensors $\mathsf{S}_1, \mathsf{S}_2, \mathsf{S}_3$ yield a measure of T according to the following linear relationships:

$$\begin{cases} \mathsf{S}_1 : x_1 = T + b_1 + v_1, \ v_1 \sim \mathcal{N}(0, \sigma_1^2), \\ \mathsf{S}_2 : x_2 = T + b_2 + v_2, \ v_2 \sim \mathcal{N}(0, \sigma_2^2), \\ \mathsf{S}_3 : x_3 = T + b_3 + v_3, \ v_3 \sim \mathcal{N}(0, \sigma_3^2), \end{cases}$$

where b_1, b_2, b_3 are **known constants**, and v_1, v_2, v_3 are mutually uncorrelated Gaussian processes with zero mean and known variances σ_i^2.

a) Consider a linear estimator for T of the form:

$$\widehat{T}(x_1, x_2, x_3) = \alpha x_1 + \beta x_2 + \gamma x_3 + \delta,$$

where $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ are parameters to be determined.

Devise a set of constraints on $\alpha, \beta, \gamma, \delta$ so that the estimator $\widehat{T}(x_1, x_2, x_3)$ is unbiased.

[3 Marks]

b) Write an expression for the variance $\text{var}[\widehat{T}]$ of the estimator in terms of $\alpha, \beta, \gamma, \delta, \sigma_1^2, \sigma_2^2, \sigma_3^2$.

[4 Marks]

c) Compute the values $\alpha^\circ, \beta^\circ, \gamma^\circ, \delta^\circ$ of $\alpha, \beta, \gamma, \delta$, respectively, for which $\text{var}[\widehat{T}]$ is minimised, while preserving unbiasedness of the estimator.

[4 Marks]

d) Suppose that a large number N of measurements $\{x_1(i), x_2(i), x_3(i)\}_{i=1}^N$ are collected from the three sensors $\mathsf{S}_1, \mathsf{S}_2, \mathsf{S}_3$ and consider the average estimator

$$\widehat{T}_N = \frac{1}{N} \sum_{i=1}^{N} \widehat{T}^\circ(x_1(i), x_2(i), x_3(i)),$$

with $\widehat{T}^\circ(x_1, x_2, x_3) = \alpha^\circ x_1 + \beta^\circ x_2 + \gamma^\circ x_3 + \delta^\circ$, where $\alpha^\circ, \beta^\circ, \gamma^\circ, \delta^\circ$ are the parameters determined in your answer to Question b).

i) Show that $\widehat{T}_N$ still is an unbiased estimator for T.

[3 Marks]

ii) Compute $\text{var}[\widehat{T}_N]$ and ascertain whether $\lim_{N \to \infty} \text{var}[\widehat{T}_N] = 0$.

[3 Marks]

iii) Compute (if possible) the minimum sample size $N_{\min}$ such that

$$\text{var}[\widehat{T}_{N_{\min}}] < 0.02$$

[3 Marks]